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REVIEW ARTICLE

A review of adaptive equipment and technology for exercise and sports activities for people with disabilities

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ABSTRACT

Individuals with disabilities face psychological, environmental, and resource-related barriers to participating in exercise and sports. The lack of participation in exercise observed in the disability community poses great concern to this population's health. This review discusses commercially available adaptive equipment and technology for exercise and sports that help people with disabilities circumvent barriers to exercise. The methods section highlights various categories of adaptive tools and their cost, accessibility, ease of learning, and progress level to help people with disabilities determine points of entry to fitness that align with their needs. Additionally, fitness-related businesses, fitness device developers, and researchers can leverage the discussed findings to understand gaps in this field to further advance adaptive equipment and technology and help facilitate widespread use. The paper serves as a comprehensive resource to researchers, entrepreneurs, and consumers to guide developing, accessing, and marketing adaptive exercise technology.

> IMPLICATIONS FOR REHABILITATION

- A majority of people with disabilities experience barriers to participating in exercise and sports which can drastically affect their physical and mental well-being.
- Although there are commercially available adaptive equipment and technologies for exercise and sports, limited accessibility, cost, and lack of options for various disabilities can hinder participation in physical activities.
- There are many developments needed in terms of technology and programs that can address issues faced by the disability community regarding exercise and sports participation.

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KEYWORDS

Adaptive exercise; adaptive sports; adaptive exercise equipment; exercise for people with disabilities; accessible exercise; adaptive technology

Introduction

It is estimated that 16% of the worldwide population experiences a significant disability that can severely impact an individual's daily living activities and, subsequently, their ability to maintain a healthy lifestyle [1]. People with disabilities (PWD) face a higher risk of developing non-communicable chronic diseases, often due to a lack of physical activity. It is reported that nearly 60% of adults with a mobility disability engage in no aerobic activity, and those who are inactive are more likely to have at least one of four chronic conditions, i.e., heart disease, stroke, diabetes, and cancer, compared to their physically active counterparts [2]. In addition to physical health risks, individuals with disabilities, particularly those who are in the early stages of their disability, commonly experience fear, anger, depression, and anxiety [3]. Physical exercise is an essential part of maintaining both physical and psychological well-being. Regarding the physical aspect, exercise helps in regulating the anti-inflammatory response, producing reproductive and metabolic hormones, and improving immune function [4]. Without proper regulation of these necessary processes, there is an increased risk of developing cancer, obesity, and cardiovascular diseases. As for the psychological aspect, exercise has been proven to improve mental health due to the release of endorphins in the brain during exercise [5]. The development of these physical and psychiatric comorbidities in PWD will pose

added health needs, as well as decrease quality of life and lower life expectancy. Considering the positive impact exercise has on preventing disease and improving physical and mental health, it is imperative to find ways to increase participation in exercise for people in the disability community.

Reports suggest that there are many psychological, resource-related, and environmental barriers hindering physical activity among PWD [6]. Many PWD struggle with psychological issues such as low self-esteem and negative body image, which further reduce their motivation to participate in exercise activities, especially when others are present [7]. They also report feeling anxiety over the fear of injury when participating in exercise since their disability could increase safety risks during physical activity [8]. Although some non-profit organizations aim to promote fitness education for PWD, there are few efforts at a commercial level that could reach a wider audience to teach PWD how to safely participate in physical activity. For instance, fitness coaches may not be trained to provide the appropriate supervision or coaching for PWD to ensure proper form and safety, and popular fitness apps are only aimed towards the general population [9]. Therefore, programs need to be implemented to improve exercise literacy and enable more independence in workout spaces (home or gym) for the disability community. Regarding resource-related issues, many gyms do not have accessible exercise equipment or buildings, and adaptive sports are not widespread or affordable

[9, 10]. Purchasing adaptive exercise equipment for home use may be a reasonable solution to this issue. However, this equipment is generally unaffordable, with limited options to choose from [9]. Moreover, an individual's living environment, urban or rural, can impose restrictions. Some disabilities can limit self-transportation, and without assistance or accessible transit options, these individuals cannot go to gyms or sports groups they desire to be a part of. It has been reported that PWD find transportation fifteen times more troublesome than for people without disabilities [1]. Additionally, PWD looking to participate in physical activity may not even live near gyms and sports groups that have the means to accommodate them. The above-mentioned barriers to exercise for PWD complicates their ability to maintain good health, making it vital to find solutions to address them.

Objective

Multiple approaches can be taken to eliminate the barriers to exercise for the disability community. This review will explore the innovative and technological solutions that are currently available, as well as identify where the lack of appropriate resources exists. Specifically, this paper will discuss adaptive exercise equipment and accessories, adaptive exercise machines, and adaptive sports equipment, as well as relevant fitness and community apps that enhance the ability of people in this underserved population to engage in physical activity. Additionally, it will identify areas in need of improvement to equip fitness-related businesses, fitness device developers, and researchers with knowledge to help PWD gain better access to fitness opportunities, which would aid in maintaining a healthy lifestyle and reducing the development of chronic diseases.

Methods

Over the past several decades, researchers and engineers have created a better path for PWD to become more active in fitness and sports activities using innovative and technological solutions. These solutions cover a wide range of purposes such as making existing equipment more disability-friendly, providing alternatives to traditional exercise and sports equipment, as well as providing fitness education and community resources. These innovations can be categorized: (1) adaptive exercise equipment and accessories, (2) adaptive exercise machines, (3) adaptive sports equipment, and (4) fitness and community apps. A literature search was performed for each category, where each section below will detail examples of commercially available solutions and how they resolve the issues complicating participation in exercise for the

disability community. Additionally, factors to consider such as cost, accessibility, learning difficulty, and maturity of development were identified. Literature search for the topics and products discussed in this review was performed using scientific databases such as PubMed, Google Scholar, IEEE Xplore Digital Library, and Science Direct, as well as websites related to health, disability, accessibility, and companies mentioned in this paper (2015–2023).

Results

Adaptive exercise equipment and accessories

Exercise equipment and accessories such as dumbbells, jump ropes, mats, and resistance bands are easily available options for exercise due to their portability and low cost. Therefore, adapting this equipment to be usable for PWD is an inexpensive, low-effort solution. Tools and accessories can be created to enable existing equipment and machines to be safe and usable without assistance from others. Some notable products currently available are shown in Figure 1. The next section discusses many simple innovations (Table 1) that have enabled adaptive exercise for a variety of disabilities, including impaired lower extremities (wheelchair users), impaired upper extremities, and visual impairments.

Commercially available products

Wheelchair users. For wheelchair users, adaptive equipment should resolve issues such as difficulty reaching certain heights to grab and lift objects, instability during exercise, and safety from using heavy objects [9]. One adaptive equipment that resolves such issues is the Fusion Wheel - All in One Portable Wheelchair Gym, shown in Figure 1A, which utilizes a set of tension bands, pulleys, and bars that can be attached to the wheelchair so the user can independently perform exercises while sitting [11]. Since the Fusion Wheel is designed to be used while in a wheelchair, workouts using this tool do not strain the wheelchair lock, cause instability, or require transferring out of the chair. For modifying existing fitness devices, the WinTech Fixed seat is a useful accessory for installing on a seated device such as rowing machines and kayaks to provide stability while exercising or performing a sport that involves a fair amount of movement [12]. Regarding weightlifting, wheelchair users sometimes resort to using gym mats and other cushions to create a barrier on their lap while lifting to minimize injuries from dropping weights. The LapMat™ is a tool specifically designed to absorb shock and diffuse the weight of fallen objects evenly across the user's lap, reducing injury [13]. This device allows the user to lift weights freely without fear of injury or needing another individual to be present, giving more independence to the user.

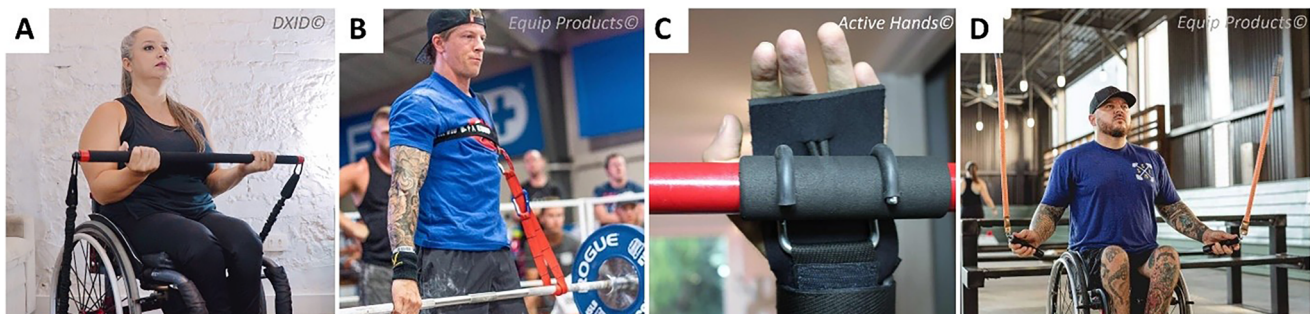


Figure 1. (A) Fusion Wheel, (B) Aldridge Arm Harness, (C) Hook Aids, and (D) Adaptive Multi Ropes.

Table 1. Examples of adaptive exercise equipment and accessories.

Product	Intended use	Adapted for	Reference
Fusion Wheel	Upper body	Wheelchair users	[11] Figure 1A
WinTech Fixed Seat	Securing upper body	Wheelchair users	[12]
LapMat™	Protection for weightlifting	Wheelchair users	[13]
Aldridge Arm Harness and Strap	Dead lifting	Single-arm individuals	[13] Figure 1B
Black Iron Master and Trainer	Grip assist for weightlifting	Single-arm individuals	[14]
The Active Hands Hook Exercise Aids	Grip assist for bars, machines with bars, handlebars etc.	Individuals with impaired hand function	[15] Figure 1C
Barbell Markers	Tactile assist for weightlifting	Visually impaired	[13]
Adaptive Multi Ropes	Cardio	Wheelchair users	[13] Figure 1D
Smart Gear Mono Rope	Cardio	Single-arm individuals	[13]
Visually Impaired White Jump Rope	Cardio	Visually impaired	[13]

Impaired upper extremity. Individuals with a single or impaired arm face obstacles during some exercises requiring two points of contact, such as weightlifting. The “Aldridge Arm” Harness and Strap™, shown in Figure 1B can be strapped to the upper body and assists in deadlift exercises by providing a second point of contact, which disperses the physical load on the body to prevent muscle injuries [13]. However, achieving upper body exercises that involve grasping or holding above the shoulders requires a single-arm user to have a prosthetic with significant weight tolerance and grip strength. Currently, there are very few commercial upper extremity prosthetics with these capabilities, as most must be custom-designed to fit the unique anatomy of the user [16]. The exception is weightlifting end-effectors with a claw-like design that attaches to a person’s existing prosthetic joint, like the Black Iron prosthetic accessory [14]. It can provide safe and secure mechanical grasping and locking while performing weightlifting and other upper-body exercises. Grasping handles can also be difficult for people who have impaired grip strength. The Active Hands® Hook Exercise aids are a set of hooks that can be placed on the hands to be used in the gym for gripping and releasing equipment, as shown in Figure 1C [15].

Visual impairment. People with visual impairment face challenges during exercise, as most equipment is designed with visual markers to ensure correct and safe use. Simple solutions to help this population exercise more safely involve modifying existing equipment with tactile or high-contrast markers. Barbell Markers made by Equip Products are large physical markers that can be placed on the barbells to be felt by the user to quickly and easily find the correct hand placement to prevent dangerous load-shifting incidents [15]. Additionally, braille markers for machine buttons and instructions can be added to existing equipment in gyms to indicate how to use the equipment properly [17].

Factors to consider

Cost and accessibility. Adaptive exercise equipment and accessories are a great option for maintaining health due to their affordability for both individuals and gyms. However, some accessories are designed to be used with existing equipment and machines. For instance, hook exercise aids are only useful when using exercise machines with handlebars, and lap mats are only used with weightlifting sets. Even though those accessories are affordable alone, they are only useful when the user has access to compatible equipment, which may be an additional cost or require access to a gym. Regardless, there are many standalone adaptive exercise equipment that can be used anywhere.

Learning difficulty. One of the biggest factors to consider for PWD looking for easy entry to fitness is that this type of equipment,

which is seemingly simple to use, may be difficult to learn (Figure 2). As most of this equipment is inherently designed for free body exercises, some users might require additional assistance or time to learn how to do exercises with proper form. Equipment should come with clear instructions on how they should be used by PWD, in the case that trained fitness coaches are not available. This learning curve can create an adoption issue for some people who are less motivated to exercise on their own. More advanced devices must be made that focus on maintaining correct exercise form and ease of use with motivation tactics.

Progress level. Adaptive exercise equipment and accessories are easily developed and can quickly penetrate the market, as they are low cost, low risk, and simple in design. However, their simplicity does not allow for much room in the design for adapting to multiple disabilities or multi-use purposes. For instance, one piece of equipment may be designed specifically for wheelchair users, but that equipment does not have the capacity to be designed for a wheelchair user who also has an impaired hand. As a result, there is room for new development of exercise equipment that is designed to be more inclusive of multiple disabilities. Currently, exercise equipment is typically designed just for one disability at a time. For instance, Equip Products has created three variations of adaptive jump ropes, each one for a different disability [13]. One is the Adaptive Multi Ropes (Figure 1D), which are designed for seated athletes to perform the upper body and core actions of jump roping without needing to move the lower body. The Smart Gear Mono Rope features one long handlebar that connects both ends of the jump rope so that it can be used by individuals with only one functional hand. There is also the Visually Impaired White Jump Rope which provides stark contrast between the floor and rope. One disadvantage to this approach, however, is that fitness centers may be reluctant to purchase this equipment due to the sheer number of options and unknown demand. There is a need for research in understanding how gyms having adaptive exercise equipment in their inventory affects membership sales, particularly, how many of those members have a disability and what their disabilities are so that they can make conscious decisions to purchase the right adaptive equipment and actively participate in creating inclusive workout environments for their local community [18].

Adaptive exercise machines

Compared to the basic equipment described in the previous section, exercise machines are more complex in their design with the intention of making exercise easier and more convenient to perform for the user [19]. Each machine is designed to be used

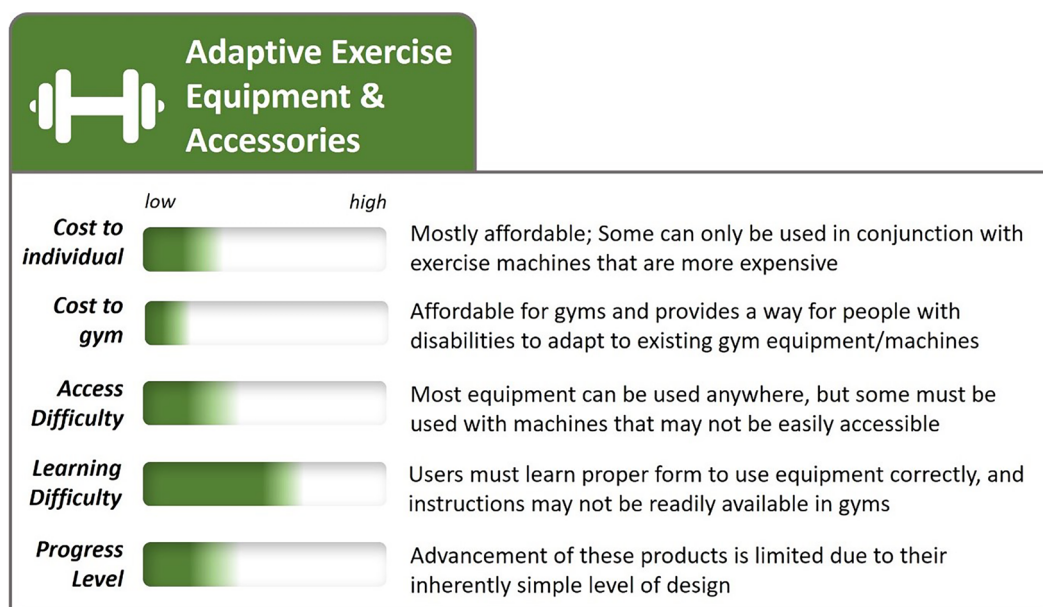


Figure 2. Adaptive exercise equipment and accessories are a great entry-level option for PWD who want to participate in fitness due to their low cost and portability; however, learning to use the equipment properly may take some additional effort.

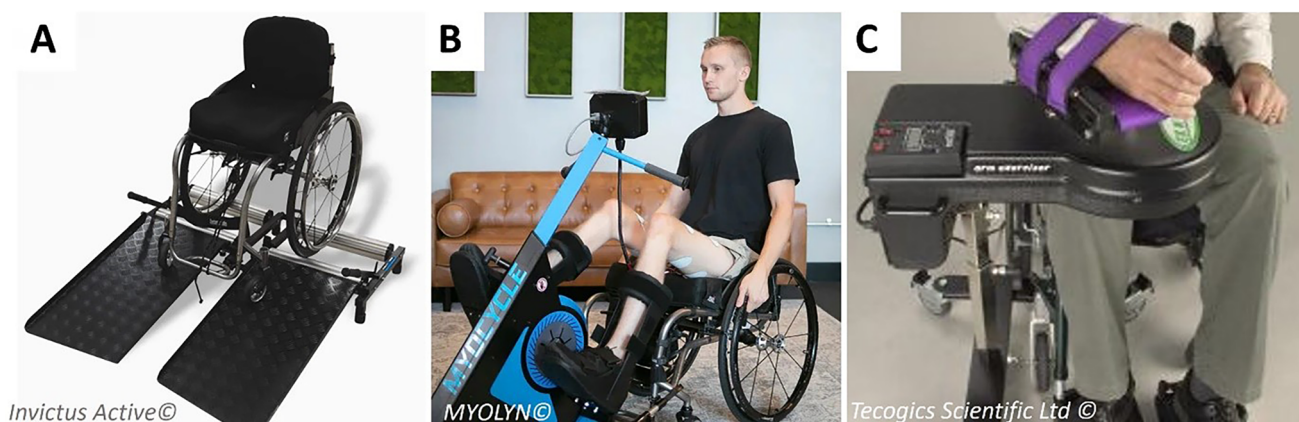


Figure 3. (A) Invictus Active Wheelchair Trainer, (B) MyoCycle, (C) Ex N' Flex EF-100.

for specific types of exercise, where it aids in positioning the body and guides the exercise with moving parts (handles, bars, pedals, etc.). However, literature shows that most of this equipment is inaccessible to people with physical disabilities, which is a major preventer of them wanting to go to the gym [18]. To overcome this issue, several companies have created adaptive exercise machines like the ones shown in Figure 3. Table 2 lists a few adaptive exercise machines which are discussed in the next section. In some cases, universal design is implemented to accommodate people with and without disabilities, while other machines are designed to be used by individuals with a specific disability.

Commercially available products

Universal design. Universal equipment that can be used by people with and without disabilities can be challenging to develop. However, there are a few exercise machine companies that have aimed to achieve this. SCIFIT is one American fitness equipment manufacturer that has developed a line of adaptive workout machines for both cardio and muscle strengthening exercises [20]. There are two main types of machines, an ergometer for hand cycling and a recumbent

stepper for low seated stair climbing, as well as a total body machine type that combines the two. Variations of the three machine types exist, and some have adaptive features including placement of handles and adjustment levers at accessible locations, removable seats for wheelchair user access, range of motion settings, low starting resistance with Iso-Strength to accommodate many strength levels, and color coded adjustments for users with visual impairment. The total body system utilizes dependent upper and lower body cranks to passively assist when a user may have more strength in one transverse half of their body than the other. Concept2 is another company that has developed several adaptive exercise machines, usable for people with and without disabilities [22]. The RowErg is a rowing ergometer that has low, wide-base seats for stability, but can also be adapted with add-ons such as fixed seats [12] and grips aids [15]. The SkiErg is another adaptive machine from their line that allows space for use while seated in a wheelchair, with the addition of handle hooks that hold the handles at a lower location on the machine for easy access. Concept2 has an app, ErgData, that is compatible with all their machines with workout tracking and voice over features for people with visual impairments. Some well-known

Table 2. Examples of adaptive exercise machines.

Product	Intended use	Adapted for...	References
SCIFIT PRO series and Recumbent Stepper	Cardio; Upper and lower body options	Wheelchair users, impaired upper and lower limb mobility, visually impaired	[20]
Concept2 RowErg	Cardio; Total body	Wheelchair users, impaired grip, visually impaired	[22]
Concept2 SkiErg	Cardio; Upper body	Wheelchair users, impaired grip, visually impaired	[22]
Peloton Bike with TalkBack	Cardio; with audio assist	Visually impaired	[23]
Elite RB Recumbent Bike	Cardio; with tactile and audio assist	Visually impaired	[24]
Invictus Active Trainer	Cardio	Wheelchair users	[25]
MyoCycle	Cardio, lower body	Impaired lower limb mobility	Figure 3A [26]
Ex N' Flex EF-100	Wrist, arm, and shoulder	Impaired upper limb mobility	Figure 3B [27]
Pro Tone™ Fitness Machine	Upper body	Wheelchair users, impaired upper limb mobility	Figure 3C [21]

exercise machine manufacturers are starting to modify their product designs to accommodate PWD. The exercise bike, Peloton®, has a feature called TalkBack for people with visual impairment, which is a screen reader that allows users to receive spoken feedback [23]. The 3G Cardio Elite RB Recumbent Bike is another machine that utilizes tactile and auditory buttons for users with visual impairment [24]. Overall, these companies have successfully adapted their equipment lines to be usable by both people with and without disabilities, and they should be viewed as a model for adapting machines to have universal design.

Specially designed. Although many exercise machines can adopt universal designs, there are some cases where the design must be substantially changed to be used by people with a particular disability. For example, the Invictus Active Wheelchair trainer, shown in Figure 3A, is a treadmill exclusive to wheelchair users [25]. Unlike traditional treadmills, the wheels of the wheelchair roll over spinning cylinders. This treadmill provides both cardiovascular and strength exercise by enabling wheelchair users to do wheelchair sprinting with the conveniences of traditional treadmill features such as indoor use and distance tracking. Another example of an adaptive exercise machine that is designed for a specific disability is the MyoCycle, shown in Figure 3B. It is a portable cycling machine for people with impaired lower extremities due to a neurological condition or disorder [26]. The exercise machine uses functional electrical stimulation to assist users to perform the cycling motion, similar to a standard cardio cycling machine. An adapted cycling machine like MyoCycle enables people with impaired lower extremities to gain the benefits of cardio exercises that improve cardiopulmonary function, muscle strength, and in some cases walking/standing ability [28]. For those who face disabilities affecting the mobility of their upper extremities, there are exercise machines that can account for limited upper body strength and contact points. The Ex N' Flex EF-100 (Figure 3C) is a muscle-strengthening machine for those with little or no control over their arm, shoulder, and trunk movement [27]. In the machine's passive mode, the user holds onto a crank, and the electrical motor rotates the arm and shoulder in a slow orbital motion. For users with more arm strength, the machine's active mode allows the user to assist the motor in moving the arm crank with the user's own muscle strength. The Aqua Creek Pro Tone™ Fitness Machine is for upper body weight-lifting workouts and designed specifically for wheelchair users, including those at varying levels of mobility and gripping capabilities. It has a designated space in the weight-lifting rack for a wheelchair to be parked in and makes changing weights easy for people with limited mobility [21]. Even though these machines would be only usable by a limited population, it is still important to develop these types of machines.

Factors to consider

Cost and accessibility. The biggest barriers to this option of fitness are their accessibility and cost (Figure 4). These machines are most often found in communal gyms, where memberships can be costly, and transportation is required. Even though gyms and fitness centers are required to comply with Americans with Disabilities Act (ADA) Standards, gyms are not legally required to have adaptive exercise machines for users of differing disabilities [29]. Gyms may be less motivated to purchase adaptive equipment because there is not enough data available on the demand to justify the high costs. Planet Fitness, with the endorsement of Paralyzed Veterans of America, the National Council on Independent Living, and the American Council of the Blind, has committed to include adaptive equipment in their gyms [30]. These fitness centers can publish statistics on adaptive equipment usage and potential return on investment for purchasing the equipment, which can encourage more gyms to accommodate PWD.

Alternatively, exercise machines can be purchased for home use, but they can cost upwards of thousands of dollars. Therefore, it is essential to find solutions to make adaptive exercise machines more widespread and affordable. One solution is expanding government assistance programs around purchasing exercise machines. Currently, Medicaid has established that assistive technology can qualify for reimbursement with a Letter of Medical Necessity, which may include some exercise equipment and machines [31]. The existence of this program may also not be well known, therefore, initiatives that aim to educate PWD and their physicians on assistance programs should emphasize this option. Lowering the barrier to financial assistance, such as reimbursement for adaptive exercise machines or gym memberships, would help to encourage fitness participation.

Learning difficulty. Although exercise machines require a higher level of engineering in their development, they are easy to learn how to use, making them a great option for people who have weaker exercise literacy. They are typically designed to position the user in the correct body form, along with moving parts that will guide movement. Many adaptive exercise machines have instructions on how to use them, either posted in print on the machine itself or built within a screen/app. Smart exercise machines that can record data such as distance, number of reps, or heart rate provide meaningful feedback to the user that can help them gauge their progress and motivate them to adhere to their exercise routine.

Progress level. Adaptive exercise machines are not widespread for both individuals and gyms. One reason may be that the majority of these machines are produced by smaller, lesser-known companies. People may be weary of ordering from independent retailers that don't have the established reputations of larger

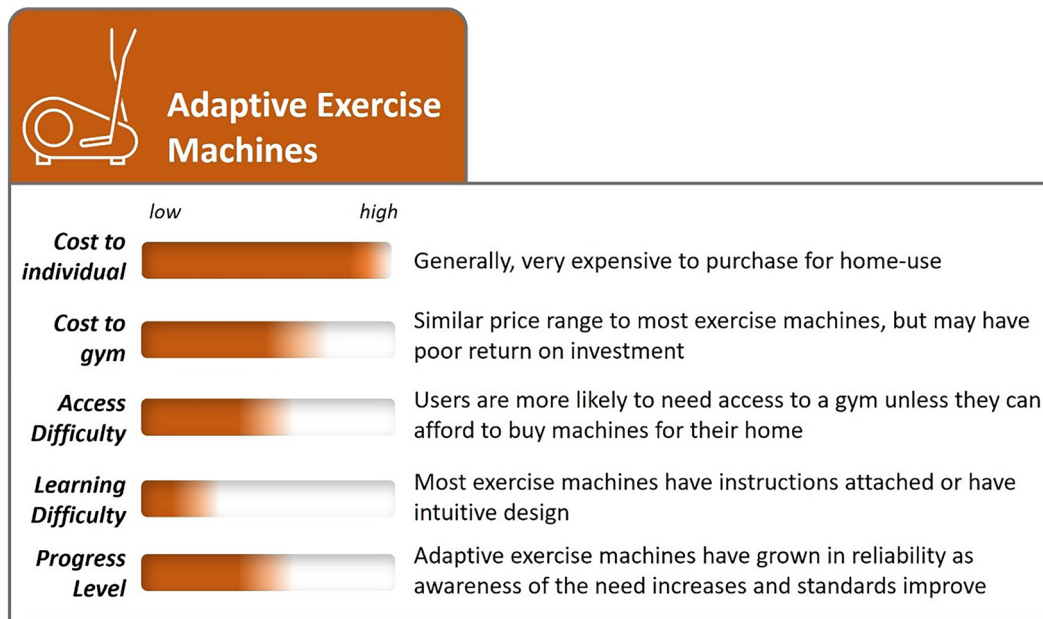


Figure 4. Adaptive exercise machines are a great entry level option for PWD who want to participate in fitness due to their ease-of-use and tracking capabilities, however, their high cost is a major barrier for both individuals and fitness facilities.

brands, which may discourage consumers from purchasing costly equipment. A way to help mitigate this issue is to have disability advocacy organizations review and endorse equipment brands. In the United Kingdom, a program called the Inclusive Fitness Initiative (IFI) has been established where their mission is to support fitness and recreation centers to become more inclusive and accessible to PWD. In addition, they can give exercise equipment and machines an “IFI Mark” accreditation, which confirms that the product meets their standards [32]. This foundation has set a precedent for similar groups to follow in other countries, including the United States (US). The IFI and a company called Beneficial Designs are working together on research funded by the National Institute on Disability and Rehabilitation Research to create the American Society for Testing and Materials (ASTM) guidelines that will ensure the quality of adaptive fitness machines and equipment in the US [33]. Although the ASTM provides inclusive fitness standards for the US, an accreditation system that is prevalent like the IFI Mark is yet to be established. The Rehabilitation Engineering and Assistive Technology Society of North America (RESNA) is also making great strides towards developing standards such as guidelines and technical reports in order to establish inclusive access marks for fitness facilities and equipment so that PWD can identify what is meeting accessibility requirements [34]. Furthering research on adaptive exercise machines can help to understand the effectiveness of these products on improving the overall health of PWD, as well as establish their efficacy as a tool for increasing exercise participation. Data from those studies would help increase marketing and consumer awareness of these technologies.

Adaptive sports equipment

Adaptive sports have enabled PWD to participate in their favorite sports and recreational activities through the development of specialized equipment and modified game rules to accommodate different physical and cognitive limitations. Sports equipment such as balls and hitting sticks, as well as mobility devices, like the

examples shown in [Figure 5](#) have been modified to be used by athletes with disabilities for participating in adaptive sports. To date, over 70 sports have been adapted for PWD so they may participate or compete in team sports (basketball, soccer, and volleyball, etc.), as well as solo sports (golf, handcycling, fencing, etc.) and outdoor recreation (rock climbing, kayaking, skiing, etc.) [35, 36]. Playing adaptive sports is a great way to encourage regular participation in exercise and to build community among athletes with similar disabilities. Sports leagues also provide an excellent support network of coaches and teammates to ensure proper training, safety, and community engagement. There are many adaptive sports equipment available, and [Table 3](#) shows several examples which are reviewed in this section covering three areas of different sports interests: entry-level sports equipment, outdoor recreation sports equipment, and competitive sports equipment.

Commercially available products

Entry-level sports equipment. For PWD that are new to sports or only have interest in playing casually, there are many simple adaptations to existing equipment that can aid them in playing sports. It should be noted that there are many sports for PWD to participate in which may not require adaptive sports equipment at all. Some examples include disc golf, pickleball, and seated volleyball [35]. Those sports serve as a low barrier to entry since no special equipment is required, and they can be played with people of all abilities. Regarding adaptive equipment, tools that are used for hitting, throwing, and swinging (balls, pucks, sticks, bats, etc.) can be easily modified for adaptive sports, and their compactness makes them easy to use and transport. Audible balls are used in many sports for people with visual impairments ([Figure 5A](#)). They typically contain bells or make beeping sounds when the ball is in motion, which allows players with visual impairments to be able to track the ball [37]. The FlagHouse Overcomer is an apparatus designed to allow users with limited mobility to participate in lightweight sports by aiding them in swinging, throwing, and striking [38]. For ball rolling sports like

bowling and bocchia, assistive devices can be used for people with mobility disabilities to move the ball, such as a ramp, ball pusher (manual and electric), and a pointer for guiding ball direction [39, 40]. These types of equipment are great for entry-level users to not only acclimate to sports activities, but to also enable them to play with their families and peers.

Outdoor recreation sports equipment. For people who prefer to become more skilled at a sport at their own pace, many outdoor recreation sports are available which can be done with others or solo. Depending on the type of sport, some activities require more complex and expensive equipment to enable PWD to participate, while others are more approachable. Archery, for instance, was easily adapted due to the simplicity of the existing equipment. A variety of special bow types have been developed for different disabilities including people with upper body limitations and wheelchair users that enable shooting with the mouth and bow stability [41]. In contrast, sports like adaptive cycling and skiing are modified from already costly equipment, making them more of an investment for serious hobbyists. There are many types of bicycles used in cycling to accommodate different disabilities [42]. Handcycles for those with lower limb mobility impairments can be powered through arm movements. Recumbent bicycles are set lower to the ground, creating more stability than a traditional bicycle. Tandem bikes that have two seats enable those with visual impairments to ride, so one sighted rider may guide the bicycle while both riders pedal. In adaptive skiing, there are ski options which similarly accommodate the disabilities mentioned for cycling [43]. A mono-ski has a molded seat low to the ground for people with lower limb mobility impairments to balance on a single wide ski. The tri-ski has a platform of three skis and handlebars for steering to provide better support and control for people with impaired balance and coordination. For people with visual impairments, a tandem ski is also available to enable riding while a second person stands in the back and steers. Adaptive equipment for outdoor recreation sports are key to enabling PWD to participate in activities when access to other players or an adapted playing field/court is limited.

Competitive sports equipment. Although many sports can be played casually for people without disabilities, this remains a barrier for PWD who may want to participate in adaptive sports like basketball, soccer, or golf that require more complex assistive devices. With the need for expensive and customized equipment, PWD who participate in these sports are typically investing in playing on an organized team or at a more competitive level. Various companies create wheelchairs that are

specifically designed for adaptive sports that require speed and mobility such as basketball, tennis, and baseball. Those wheelchairs come with ratchet straps, rear wheels that are angled for better stability, and particular frame materials to improve speed and turning [44]. Some sports require the use of durable electric wheelchairs, called powerchairs, which are inclusive for wheelchair users with limited upper mobility that cannot easily roll a chair themselves [45]. In powerchair football, the chairs are equipped with a square metal frame around the feet to kick and block an over-sized soccer ball. Power hockey uses the same type of chair where the users play an adapted form of hockey in an indoor court. Adaptive hockey (Figure 5B) also has a version that can be played on a real ice rink, where the mobility device is a seated sled with two hockey skate blades for gliding on the ice [46]. There are other mobility devices, such as adaptive golf carts (Figure 5C), which not only transport the player across the field, but also allow the player to stay supported in an upright position while they swing the golf club [47]. In certain adaptive sports, especially at the professional level, the use of personalized equipment is more common. This is because adaptations are fit to each athlete's specific disabilities, as not every athlete will have the same mobility limitations, or the athlete may need equipment custom fit for their body. For example, track athletes with amputated lower limbs can use running blades to be involved in competitive track and field [48]. The blades are usually custom-made or have options to adjust the height and fit. These adaptive equipment enable PWD to participate in sports with greater aspects of competition and demand for athletic skill.

Factors to consider

Cost and accessibility. Adaptive sports are a highly effective way to encourage participation in exercise for the disability community but cost for the equipment is a major barrier (Figure 6). While some equipment, like adaptive soccer balls and volleyballs, are fairly inexpensive, adaptive wheelchairs and running blades can cost thousands of dollars. For people who are only interested in casual sports or entry level leagues, purchasing this equipment may not be a good investment. Gyms for adaptive fitness and sports could aim to provide loaner equipment to reduce the financial burden for those who want to play casually, but this would be an expensive investment with uncertainty regarding the benefit-cost ratio. On top of equipment costs, some sports leagues charge membership fees that can also be hundreds of dollars annually. If players decide to play in more competitive or professional leagues, the associated costs of traveling to games and receiving professional training can also be a major financial burden. Therefore, continual efforts should be made to reduce the cost of adaptive sports equipment, as well as provide support



Figure 5. (A) Audible Goalball, (B) Adaptive Hockey Sled and Sticks, (C) SoloRider Adaptive Golf Car.

Table 3. Examples of adaptive sports equipment.

Product	Intended use	Adapted for...	References
Audible Sports Balls	Basketball, volleyball, football/soccer, goalball	Visually impaired	[37] Figure 5A
The FlagHouse Overcomer	Swinging, throwing, striking	Individuals with upper mobility impairment	[38]
Ball Ramps, Pushers, and Pointers	Bowling, boccia	Individuals with upper/lower mobility impairment	[39, 40]
Archery Bows	Archery	Individuals with upper mobility impairment	[41]
Handcycles	Cycling	Individuals with lower mobility impairment	[42]
Recumbent Bicycles	Cycling	Circulation, musculoskeletal, and/or balance impairments	[42]
Tandem Bicycles	Cycling	Visually impaired	[42]
Mono-ski	Skiing	Wheelchair users	[43]
Tri-ski	Skiing	Circulation, musculoskeletal disorders, and/or balance impairments	[43]
Tandem Ski	Skiing	Visually impaired	[43]
Sports Wheelchairs	Multi-sport	Wheelchair users	[44]
Power Wheelchair	Football, soccer, hockey	Wheelchair users	[45]
Adaptive Hockey Sled and Sticks	Hockey	Wheelchair users	[46] Figure 5B
Adaptive Golf Carts	Golf	Lower mobility, circulation, and/or balance impairments	[47] Figure 5C
Running Blades	Running	Lower limb amputees	[48]

**Figure 6.** Adaptive sports is a well-established global community that has many opportunities from entry level sports to olympic level. While adaptive sports equipment enable access to engaging in exercise with great community support, high costs for both equipment and participation are major barriers.

to disabled sports programs. In the US, there are several adaptive sports organizations that lead the effort in promoting adaptive sports for children and adults with disabilities by hosting sporting events, funding travel to competitions, and implementing adaptive sports programs in schools [36]. These groups have done substantial work in fostering community, support, and fair organization among the many existing adaptive sports.

Learning difficulty. Adaptive sports can be played by anyone from casual play to entry level leagues to Paralympics. In casual settings, learning to play sports can be fun and easy, however, there may be difficulties in learning how to play with adaptive equipment if a trained person is not involved. In sports leagues, a person will have to put in a lot more effort to learn how to play a sport well especially if they are competing with others or are playing on a team. The pressure to become better within a sport community can possibly lead to demotivation or require the person to dedicate a lot of time, effort, and energy to improve as an athlete.

Progress level. Adaptive sports equipment have progressed well, as the community is heavily supported by disability community organizations and philanthropists, as well as people who were athletes before acquiring their disability. As a result, adaptive sports equipment have been continually evolving and developing for many years. This area of assistive technology still has many opportunities to advance and is positioned to remain that way due to the popularity of sports. One recent study described the significant role that sensing and training technologies have in enabling review of performance for athletes to improve their skills and strategy. They also identified that these technologies are underdeveloped for the adaptive sports community and how they could be highly beneficial as training aids, health trackers for expensive equipment like wheelchairs, or smart equipment that track athletes' performance metrics. To enable development of those technologies, more adaptive sports data need to be collected, such as kinematic models and performance and health metrics of disabled athletes [49].

Fitness and community apps

Fitness and community apps are online programs that can help individuals increase their knowledge and participation in exercise and sports. The benefit to apps, using the modern use of the term as a downloadable software package for a mobile device or computer, is that they are easy to find and access through an online app store. Using these apps can help PWD overcome resource-related, environmental, and psychological barriers to exercise by providing fitness education and safety, creating tailored exercise routines, finding accessible gyms, connecting PWD who are interested in fitness, and providing emergency services and mental health tools. Many in the disability community lack the resources to participate in exercise, whether that is due to inability to participate in traditional workout routines, lack of transportation, difficulties finding disability-friendly gyms, or just feeling social isolation during exercise. Many apps are available, such as those shown in [Figure 7](#), that can be utilized to tackle these barriers. The apps listed in [Table 4](#) will be detailed in the following sections to discuss how they can be used and combined to tailor a fitness lifestyle that is accessible and motivational, as well as safe and supported by community and technology. These apps can be categorized by their main purposes: fitness, accessible gym finders, community and social support, and safety and pain management.

Commercially available products

Fitness. Several fitness apps are available that provide tutorials on workouts adapted to accommodate different levels of physical abilities. For instance, Evolve21 is an exercise app that provides users of all abilities with workouts for cardio, yoga, and meditation programs. The app can provide modified exercises for individuals with different disabilities [50]. Accessercise is another app for people with impairments to not only learn about adapted exercises, but they can also build their own workout routines, view a directory of accessible fitness centers, and interact on their social hub to connect with others for support and motivation [51]. People with cognitive disabilities may face difficulties in participating and adhering to fitness routines due to challenges in focusing, triggering social settings, and fast-paced learning environments [62]. Let's Go Fitness is an app developed for people with cognitive disabilities to access a virtual fitness community so that they can exercise with people at their fitness level, as well as learn exercises from people who are trained to teach those with cognitive disabilities [52]. For wheelchair users, the Wheel With Me Adapt Fit app has workouts specifically designed to consider their functional mobility limitations [53]. These apps are an entry-level resource for PWD to learn how to engage in exercises without fear of injury and to track progress.

An additional advantage to fitness apps is that they can be designed to integrate with exercise equipment or machines to increase motivation to exercise by enabling users to track their progress. For example, the BitGym app is designed to connect to a cardio machine to provide virtual tours of different locations while tracking distance and speed, as shown in [Figure 7A](#). BitGym has been programmed to accommodate the adaptive treadmill Invictus Active Wheelchair Trainer ([Figure 3A](#)), so that wheelchair users can benefit from this interactive fitness technology [54]. Apps like BitGym, can serve as a guide on how to connect the technology of apps to different physical exercise tools to make participation in physical activity fun and motivating.

Accessible gym finders. There are apps designed to help locate places like gyms and sports facilities that have proper accommodations for those with disabilities. AccessNow is an app

for PWD that catalogs accessible locations worldwide with information such as the availability of wheelchair ramps, accessible parking spots, braille signs, and wheelchair-friendly restrooms [55]. iAccess Life is another app that lets users with disabilities share reviews and research places to determine their accessibility [56]. Users can utilize these apps to locate gyms that cater to PWD and provide adaptive exercise equipment.

Community and social support. Community apps are an important tool for PWD to meet others with shared experiences, connect with peers who have similar fitness and sports interests, and find emotional support in their fitness journeys. The Making Authentic Friendships app is one such tool where PWD can find friends with similar interests, ages, diagnoses, and hobbies [57]. Although the app is not specifically geared towards fitness, it can be used as a tool to connect PWD interested in fitness and sports ([Figure 7C](#)), as well as can lead to creating fitness and sports groups local disability communities. Another social connection app geared towards connecting PWD is the Australian Alvie App. Notably, membership fees for this app can be waived using Australia's National Disability Insurance Scheme due to its usefulness in improving mental wellbeing [58]. These apps could help foster friendships and connections amongst those with disabilities so they can find the social support required to begin and maintain exercise routines.

Safety and pain management. Many PWD struggle with fear of injury or pain during exercise. Apps have been developed to build emergency plans and manage pain. As shown in [Figure 7B](#), the In Case of Emergency Medical Reports app allows for storing personal and medical information so that first responders and healthcare professionals can access the user's health information, medications, and other information about their disability that is needed to help in an emergency [59]. These features give users the peace of mind to exercise independently. There are also apps aimed towards managing pain during exercise. For example, the Curable app uses a virtual pain coach that sends the user messages, activities, resources, and exercises tailored to their disability and pain needs [60]. Overall, these apps for safety and mental well-being can be critical in helping to ease psychological barriers to exercise.

Factors to consider

Cost and accessibility. As many people use mobile devices, apps are very accessible and can be used almost anywhere. Many of these apps are free or can be downloaded for a very low cost ([Figure 8](#)). Some apps may require a subscription that can become costly if the person uses it for an extended period. However, those costs can be controlled, as users can create habits and learn practices from the apps until they are confident without them. These apps are a great entry-level option for PWD to start participating in fitness due to their accessibility and affordability. Similar to the Australian Alvie App, insurance options to cover membership fees for these community and fitness apps should be highly considered in the US and other countries.

Regarding apps that are designed for exercise equipment and machines, the added cost of the equipment may pose affordability challenges. There are very few adaptive exercise machines that work with fitness apps. However, as more adaptive exercise machines continue to develop, gyms can look to purchase ones that connect with a fitness app so their members can utilize them. Gyms may also be inclined to create their own app for their

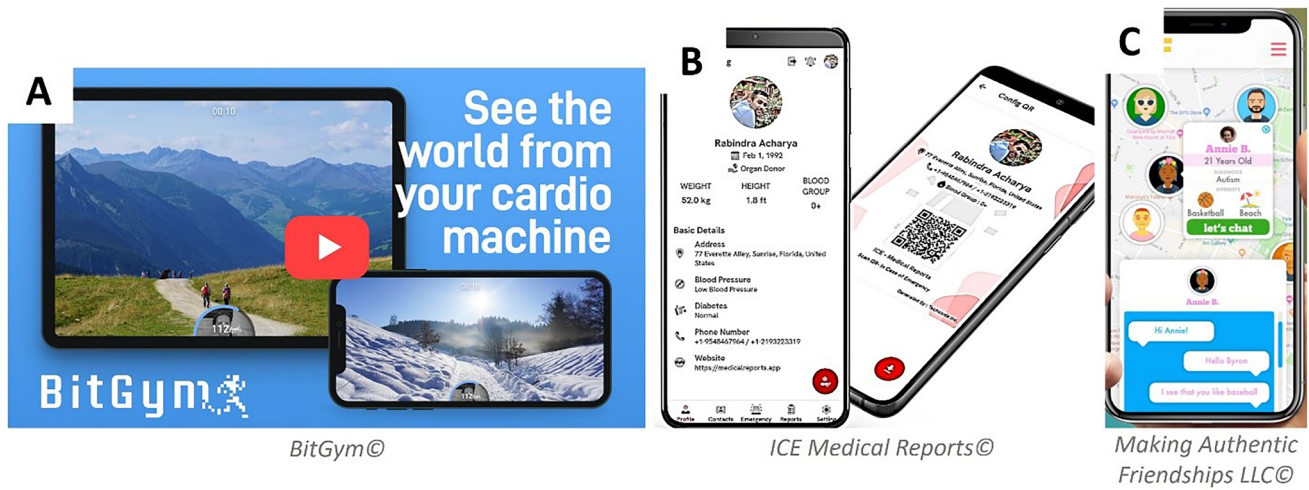


Figure 7. (A) BitGym App, (B) ICE Medical Reports App, (C) Making Authentic Friendships App.

Table 4. Examples of fitness and community apps.

Product	Intended use	Adapted for...	References
Evolve 21	Fitness, Yoga, Meditation, Community	Cerebral palsy, physical disabilities, wheelchair users	[50]
Accessercise	Fitness, Community	Physical disabilities	[51]
Let's Go Fitness	Fitness, Community	Cognitive disabilities	[52]
Wheel With Me Fitness	Fitness	Wheelchair users	[53]
BitGym	Cardio	Wheelchair users, physical disabilities	[54] Figure 7A
AccessNow	Community	All disabilities	[55]
iAccess Life	Community	Physical disabilities	[56]
Making Authentic Friendships (MAF)	Community	Cognitive disabilities	[57] Figure 7C
Alvie	Community	All disabilities	[58]
In Case of Emergency (ICE) – Medical Reports	Safety	All disabilities	[59] Figure 7B
Curable	Pain Management	All disabilities	[60]
TruFit	Fitness, Community	All disabilities	[61]

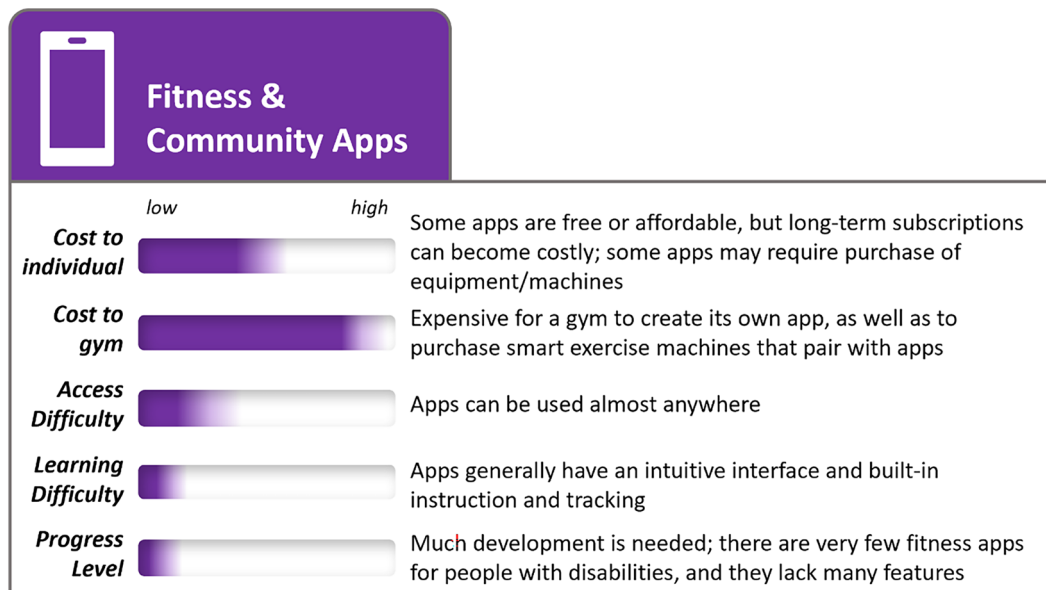


Figure 8. Fitness and community apps provide resources for PWD to easily engage in fitness due to their accessibility and ease of learning, however, they have potential to be costly in the long-term due to subscription fees. Most existing fitness apps for PWD are also underdeveloped.

members to track progress and connect with smart exercise machines. However, that can be extremely costly to develop and maintain.

Learning difficulty. Most apps are intuitive to learn how to use. However, these apps may be difficult to learn how to use for people who are not familiar with technology or mobile apps. Developers for disability-friendly apps should greatly consider including options for people with visual and hearing impairments when using a mobile device. Learning how to perform exercises through video and audio instructions may not be as helpful compared to in-person training, where an individual can have real-time and physical interaction with a trainer or peer.

Progress level. In terms of fitness apps, there are few specifically designed for PWD like Evolve21 and Accessercise. Options are still very limited, and their quality is not at the same level as fitness apps that are made for the general population. For instance, there are many apps for the general population where users can be part of an online workout group with a dedicated mentor/coach and support groups for engaging with each other. This concept of virtual fitness has been shown to be highly effective in improving physical and mental well-being [63]. There are a few adaptive fitness centers and organizations such as Disability Partnerships, Adaptive Training Academy, and Oak Hill Adaptive Sports and Fitness that host regular virtual adaptive exercise classes through videoconference platforms (Zoom, Facebook live, etc.). Although these programs are effective and fill a gap in the industry, having them on apps available on common systems like iOS and Android would be wider-reaching and provide instant access to a centralized source of fitness learning materials and programs. A recent app called TruFit has implemented a group exercise class concept that is tailored for PWD [61]. Development of more apps with group exercise programs for PWD could have a significant impact on the community and should be greatly considered. Regarding apps that can help users find accessible gyms and availability of adaptive exercise equipment, current apps can help users find accessible facilities but do not provide information on available equipment. There are apps for PWD to facilitate social interactions, however, these apps are not focused on facilitating connecting users with similar interests in exercise and sports. Overall, the apps discussed in this section each have functions that can be useful, however, more development and additional functionality is needed to enhance engagement of physical activities among PWD. Moreover, there is growing interest in integrating apps with exercise equipment and machines to improve motivation and adherence. This concept has not been well-explored with adaptive exercise equipment. The only commercial example found was the integration of the Invictus Wheelchair Trainer with the BitGym app.

Discussion

Previous literature has established the importance of exercise in the disability community, identifying exercise as a vital component to maintaining physical and mental health while preventing the development of comorbidities to existing health conditions. Efforts have been made to increase participation in exercise for the disability community, which has been greatly facilitated by the development of adaptive tools. This review looked at commercially available adaptive exercise equipment and machines, adaptive sports equipment, and fitness and community apps to examine

what remaining issues pose barriers to exercise for the disability community.

Adaptive exercise equipment and accessories were reviewed as a simple and affordable option for PWD, where many equipment can be used at home, and accessories can enable using current exercise equipment in gyms. However, these equipment can be difficult to learn how to use since they are free body exercise tools. To make adaptive exercise equipment more approachable, developers of these tools should greatly consider ways to enhance exercise literacy, such as supplementary apps and video tutorials, so that users can feel confident using them. *Adaptive exercise machines*, on the other hand, are an effective source for exercise as they are easier to use and can assist in performing motions that would otherwise be difficult for people with physical impairments. Despite their advancements in making exercise more accessible, adaptive exercise machines are expensive and not commonly found in gyms. Developers should increase their efforts in working with advocacy groups that aim to increase accessibility in gyms so adaptive exercise machines can become more widespread. For PWD that enjoy sports, many *adaptive sports equipment* are available and do not require much development. However, some of the more complex sports equipment can be very costly, making adaptive sports difficult to participate in, especially for casual and entry-level players. Gyms that can accommodate adaptive sports should aim to provide loaner equipment so that people can have better access to playing sports without having to invest in expensive equipment. In contrast, *fitness and community apps* are an affordable and accessible tool for individuals to start participating in exercise. However, there is much room for progress in adaptive fitness apps, as not many exist that can connect with adaptive exercise equipment to track progress. Many apps also lack the unification of valuable features such as accessible gym finders, virtual group exercise, and live interaction with trainers.

Despite the level of innovation that exists on the market today, there is still a significant part of this population that does not participate in any form of exercise. Looking at the adaptive tools that currently exist, several overarching conclusions can be made. For complex adaptive devices, like exercise machines and some sports equipment, finding ways to reduce the financial barrier of accessing products is essential. This can be done by reducing production costs associated with materials and labor. Many of the products discussed in this paper have been sourced from smaller businesses that do not have the capital to produce larger quantities of products. Additionally, many of these small companies do not have product liability insurance, which may not only detract development of newer technologies for the disability community, but also drive up the cost of the equipment. Aligning with larger manufacturers to do research will be a potential way to help produce equipment that costs less. For all these adaptive solutions, research on how to better market these tools is also essential. Partnering with disability organizations and creating government campaigns to showcase the types of adaptive solutions available can help make these products more widely known. Similarly, marketing the benefits of exercise for the disability community will help encourage using these products to increase exercise participation. Also, to address issues regarding transportation, the development of a disability-friendly ride-share service can help ease the burdens. Both Uber and Lyft offer Wheelchair Accessible Vehicles (WAV), but they are not always reliable, and users report unequal service [64]. This paper has been limited to include only products that are currently available on the market. Identifying ongoing research projects can also help narrow down

existing gaps in the adaptive exercise field. Furthermore, there is no consumer data on how well-known these commercial products are to the public. Performing consumer surveys is also a necessary step to direct marketing and educational health campaigns to make these adaptive solutions become more utilized within the disability community. Overall, guiding research initiatives toward improving existing products, developing new technologies, and promoting outreach programs will yield great changes for the disability community.

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